

01

WHAT IS ROS2?

MIDDLEWARE

- It is a middleware connecting the code together
- Using it shapes the architecture and how code communicates to together
- Large community with many contributions - famous in the robotics community
- Powerful libraries
- Supports both python and c++
- Allows you to use a tool called **Rivz2** that visualizes what the robot see and how the robot is functioning
- Allows you to use simulators like Gazebo to simulate the robot working!



02



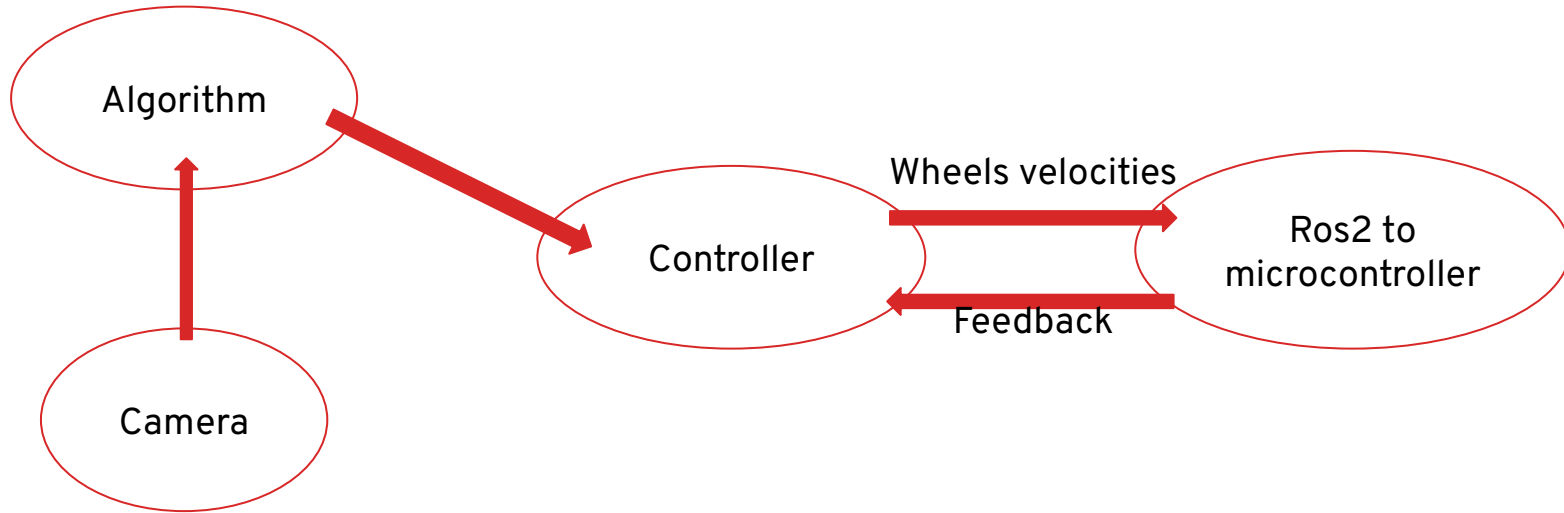
WHY ROS2?

WITH VS WITHOUT ROS

Architectural Metric	Monolithic Architecture	ROS 2 Architecture
Process Isolation	Single process / Multithreaded	Isolated Multi-process nodes
Data Exchange	Manual - gets messy if not careful	Typed Pub/Sub topics & Services
Failure Handling	Process crash kills whole robot	Node crashes; remainder stays active
Language Polyglot	Single language	C++ & Python seamlessly integrated
Real-Time Capability	Custom RTOS or cyclic loops	Native real-time executors & zero-copy

ROS2 EXAMPLE

Think of ROS 2 as a massive collection of mini-programs that talk to each other to make a robot move, see, and make decisions.





03
ROS2

STRUCTURE

WORKSPACE

A **Workspace** is simply your master project folder. It holds all your individual packages in one place so ROS2 can find, build, and run them together.

Conventionally ros2 packages are created inside a folder called **src** in the Workspace:

```
ros_ws/  
├── README.md  
├── src/      → where your packages are  
└── pixi.toml // or any extras
```

PACKAGES

- A package is basically a neatly organized folder containing node(s) - mini-program(s) - that needs to run.
- The code in the packages are nodes, all nodes in a package should achieve the mini-program the package represents
- It is either a Python package or a C++ package
Btw: the workspace can have mixed packages - some packages in c++ and others in python!

PACKAGE.XML

Think of `package.xml` as the package's ID card and instruction manual for ROS2. It tells the system:

- **Identity:** What is the package's name, version, and author?
- **License:** Is it open-source?
- **Dependencies:** What *other* packages or software does this code need to work?

(It is there in both Python and C++ packages)

PYTHON PACKAGES

```
my_python_pkg/  
├── package.xml  
├── setup.py  
├── setup.cfg  
├── resource/  
├── my_python_pkg/  
│   └── my_script.py
```

- ← The ROS2 ID card
- ← The Python installation recipe + end points
- ← Where the installed files should go (NO TOUCH)
- ← ROS2 tracking of packages(NO TOUCH)
- ← Your actual Python code folder

C++ PACKAGES

```
my_cpp_pkg/  
├── package.xml  
├── CMakeLists.txt  
├── src/  
└── include/
```

- ← The ROS 2 ID card
- ← The compiler instructions
- ← Your .cpp code
- ← Your .h header files



THANK

YOU